Exercise J-M

1. Let $f(x)=\begin{cases} x^2\sin\left(\frac{1}{x}\right) & , \quad x\neq 0 \\ 0 & , \quad x=0 \end{cases}$

Then at $x=0$: **[JEE(Main)Jan.2023]**

(A) f is continuous but f' is not continuous (B) f and f' both are continuous

(C) f is continuous but not differentiable (D) f' is continuous but not differentiable

माना $f(x)=\begin{cases} x^2\sin\left(\frac{1}{x}\right) & , \quad x\neq 0 \\ 0 & , \quad x=0 \end{cases}$ है।

तो $x=0$ पर : **[JEE(Main)Jan.2023]**

(A) f संतत है परन्तु f' संतत नहीं है (B) f तथा f' दोनों संतत है

(C) f संतत है परन्तु अवकलनीय है (D) f' संतत है परन्तु अवकलनीय है

Ans. A

Sol. at x= 0

$$\text{LHD}=\lim_{h\to 0^+}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to 0^+}\frac{-h^2\sin(1/h)}{-h}=\lim_{h\to 0}\frac{\sin(1/h)}{(1\setminus h)}=0$$

$$\text{RHD}=\lim_{h\to 0^+}\frac{f(0+h)-f(0)}{h}=\lim_{h\to 0^+}\frac{h^2\sin(1/h)}{h}=\lim_{h\to 0^+}\frac{\sin(1/h)}{1/h}=0$$

$\Rightarrow$ f(x) = is continuous and differential at $x=0$

$$\text{Now } f'(x)=\begin{cases} 2x\sin(1/x)-\cos(1/x), & x\neq 0 \\ 0, & x=0 \end{cases}$$

clearly f'(x) is discontinuous at $x=0$

2. If the function $f(x)=\begin{cases} (1+|\cos x|)\frac{\lambda}{|\cos x|}, & 0<x<\frac{\pi}{2} \\ \mu & , \quad x=\frac{\pi}{2} \\ \frac{\cot 6x}{e^{\cot 4x}} & , \quad \frac{\pi}{2}<x<\pi \end{cases}$ is continuous at $x=\frac{\pi}{2}$, then $9\lambda+6\log_e\mu+\mu^6-e^{6\lambda}$ is

equal to : **[JEE(Main)Jan.2023]**

यदि फलन $f(x)=\begin{cases}(1+|\cos x|)\dfrac{\lambda}{|\cos x|}, & 0<x<\dfrac{\pi}{2}\\ \mu, & x=\dfrac{\pi}{2}\\ \dfrac{\cot 6x}{e^{\cot 4x}}, & \dfrac{\pi}{2}<x<\pi\end{cases}$ $x=\dfrac{\pi}{2}$ पर संतत है तो $9\lambda+6\log_e\mu+\mu^6-e^{6\lambda}$ बराबर है

[JEE(Main)Jan.2023]

(A) 8 (B) 10 (C) $2e^4+8$ (D) 11

Ans. Drop

Sol. $f\left(\frac{\pi}{2}+\right)=x-\frac{\pi}{2}+e^{\frac{\cot 6x}{\cot 4x}}=\lim x-\frac{\pi}{2}+e^{\frac{\sin 4x}{\sin 6x}\cdot\frac{\cos 6x}{\cos 4x}}$

$=\lim x\to\frac{\pi}{2}+e^{\frac{4\cos 4x}{6\cos 6x}}=e^{\frac{2}{3}}$

$\text{LHL}\,f\left(\frac{\pi^-}{2}\right)=\lim_{x\to\frac{\pi^-}{2}}(1+|\cos x|)^{\frac{\lambda}{\cos x}}=e^{\lambda}\Rightarrow\lambda=\frac{2}{3},\mu=e^{\frac{2}{3}}$

So, $9\lambda+6\ln\mu+\mu^6-e^{6\lambda}$

$=9\left(\frac{2}{3}\right)+6\left(\frac{2}{3}\right)+e^4-e^4=6+4=10$

3. Suppose $f:\mathrm{R}\to(0,\infty)$ be a differentiable function such that $5f(x+y)=f(x)\cdot f(y),\forall x,y\in\mathrm{R}$. If $f(3)=320$, then $\sum_{n=0}^{5}f(n)$ is equal to : **[JEE(Main)Jan.2023]**

माना एक अवकलनीय फलन $f:\mathrm{R}\to(0,\infty)$ के लिए $5f(x+y)=f(x)\cdot f(y),\forall x,y\in\mathrm{R}$. है। यदि $f(3)=320$ तो $\sum_{n=0}^{5}f(n)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 6825 (B) 6525 (C) 6575 (D) 6875

Ans. A

Sol. $5f(x+y)=f(x)\cdot f(y)$

$5f(0)=f(0)^2\Rightarrow f(0)=5$

$5f(x+1)=f(x)\cdot f(1)$

$$\Rightarrow \frac{f(x+1)}{f(x)} = \frac{f(1)}{5}$$

$$\Rightarrow \frac{f(1)}{f(0)} \cdot \frac{f(2)}{f(1)} \cdot \frac{f(3)}{f(2)} = \left(\frac{f(1)}{5}\right)^3$$

$$\Rightarrow \frac{320}{5} = \frac{(f(1))^3}{5^3} \Rightarrow f(1) = 20$$

$\therefore 5f(x+1) = 20 \cdot f(x) \Rightarrow f(x+1) = 4f(x)$

$$\sum_{n=0}^{5} f(n) = 5 + 5.4 + 5.4^2 + 5.4^3 + 5.4^4 + 5.4^5$$

$$= \frac{5\left[4^6 - 1\right]}{3} = 6825$$

4. Let $f: \text{IR} \to \text{IR}$ be a differentiable function that satisfies the relation $f(x+y) = f(x) + f(y) - 1, \ \forall x, y \in \text{IR}$. If f'(0) = 2, then $|f(-2)|$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $f: \text{IR} \to \text{IR}$ एक अवकलनीय फलन है जो संबंध $f(x+y) = f(x) + f(y) - 1, \ \forall x, y \in \text{IR}$ को संतुष्ट करता है। यदि f'(0) = 2 है, तब $|f(-2)|$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 3

Sol. $f(x+y) = f(x) + f(y) - 1$ .....(1)

Partial differential w.r.t. x

$f'(x+y) = f'(x)$

put x = 0

$f'(y) = f'(0) = 2$

$\Rightarrow f(y) = 2y + c$

$f(x) = 2x + c$

Now put x = y = 0 in (1) we get f(0) = 1

$\Rightarrow f(0) = 0 + c$

$\Rightarrow c = 1$

$\Rightarrow f(x) = 2x + 1$

$\Rightarrow |f(-2)| = 3$

5. Let $f(x) = \left[x^2 - x\right] + |-x + [x]|$, where $x \in \mathbb{R}$ and [t] denotes the greatest integer less than or equal to t. Then, f is: **[JEE(Main)April,2023]**

(A) continuous at x = 0 and x = 1 (2) not continuous at x = 0 and x = 1

(C) continuous at x = 0, but not continuous at x = 1 (D) continuous at x = 1, but not continuous at x = 0

माना $f(x)=\left[x^2-x\right]+|-x+[x]|$ है, जहाँ $x \in \mathbb{R}$ है तथा [t] महत्तम पूर्णांक ≤ t है। तो f :

[JEE(Main)April,2023]

(A) x = 0 तथा x = 1 पर सतत है (B) x = 0 तथा x = 1 पर सतत नहीं है

(C) x = 0 पर सतत है, परन्तु x = 1 पर सतत नहीं है (D) x = 1 पर सतत है, परन्तु x = 0 सतत नहीं है

Ans. D

Sol. Here f(x) = [x(x – 1)] + {x}

$f(o^+) = -1 + 0 = -1$ $f(1^+) = 0 + 0 = 0$

f(o) = 0 f(1) = 0

$f(1^-) = -1 + 1 = 0$

∴ f(x) is continuous at x = 1, discontinuous at x = 0

6. Let f and g be two functions defined by $f(x)=\begin{cases}x+1, & x<0\\ |x-1|, & x\geq 0\end{cases}$ and $g(x)=\begin{cases}x+1, & x<0\\ 1, & x\geq 0\end{cases}$. Then (gof) (x) is :

[JEE(Main)April,2023]

(A) continuous everywhere but not differentiable at x = 1

(B) not continuous at x = –1

(C) differentiable everywhere

(D) continuous everywhere but not differentiable exactly at one point

माना दो फलन f तथा g, $f(x)=\begin{cases}x+1, & x<0\\ |x-1|, & x\geq 0\end{cases}$ तथा $g(x)=\begin{cases}x+1, & x<0\\ 1, & x\geq 0\end{cases}$ द्वारा परिभाषित है। तो (gof) (x) :

[JEE(Main)April,2023]

(A) R पर सतत है परन्तु x = 1 पर अवकलनीय नहीं है

(B) x = –1 पर सतत नहीं है

(C) R पर अवकलनीय है

(D) R पर सतत है परन्तु मात्र एक बिंदु पर अवकलनीय नहीं है

Ans. D

Sol. $f(x)=\begin{cases}x+1, x<0\\1-x, 0\le x<1\\x-1, 1\le x\end{cases}$

$g(x)=\begin{cases}x+1, x<0\\1, x\ge 0\end{cases}$

$g(f(x))=\begin{cases}x+2, x<-1\\1, x\ge -1\end{cases}$

$\therefore g(f(x))$ is continuous everywhere

$g(f(x))$ is not differentiable at $x=-1$

Differentiable everywhere else

7. Let [x] denote the greatest integer function and $f(x)=\max\{1+x+[x], 2+x, x+2[x]\}, 0\le x\le 2$. Let m be the number of points in [0,2], where f is not continuous and n be the number of points in (0, 2), where f is not differentiable. Then $(m+n)^2+2$ is equal to : **[JEE(Main)April,2023]**

यदि [x] महत्तम पूर्णांक फलन है तथा $f(x)=\max\{1+x+[x], 2+x, x+2[x]\}, 0\le x\le 2$ है। यदि [0,2] में बिन्दुओं की संख्या m इस प्रकार है जहाँ f असतत् है तथा (0, 2) में बिन्दुओं की संख्या इस प्रकार है जहाँ f अवकलनीय नहीं है, तो $(m+n)^2+2$ का मान बराबर होगा : **[JEE(Main)April,2023]**

(A) 6 (B) 11 (C) 2 (D) 3

Ans. D

Sol. $f(x)=\begin{cases}\max\{x+1, x+2, x\} & 0\le x<1\\ \max\{x+2, x+2, x+2\} & 1\le x<2\\ \max\{5,4,6\} & x=2\end{cases}$

$\Rightarrow f(x)=\begin{cases}x+2 & 0\le x<1\\ x+2 & 1\le x<2\\ 6 & x=2\end{cases}$

$\Rightarrow f(x)=\begin{vmatrix}x+2 & 0\le x<2\\ 6 & x=2\end{vmatrix}$

$\Rightarrow f(x)=\begin{cases}x+2 & 0\le x<2\\ 6 & x=2\end{cases}$

f is not continuous at $x=2$

f is differentiable in (0, 2)

$\therefore m=1, n=0$

$(m+n)^2+2=1+2=3$

option (4) is correct.

8. Let $a \in \mathbb{Z}$ and [t] be the greatest integer $\leq$ t. Then the number of points, where the function f(x)=[a + 13 sin x], $x \in (0, \pi)$ is not differentiable, is _____. **[JEE(Main)April,2023]**

माना $a \in \mathbb{Z}$ है तथा [t] महत्तम पूर्णांक $\leq t$ है। तब उन बिंदुओं, जहाँ f(x)=[a + 13 sin x], $x \in (0, \pi)$ अवकलनीय नहीं है, की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 25

Sol. $f(x) = a + [13 \sin x] \quad \because a \in I$ and $x \in (0, \pi)$

$\because$ total number of points of non-differentiability of $[p \sin x] = 2p - 1$ here $p = 13$

$\therefore$ total number of points of non-differentiability of $[13 \sin x] = 25$

9. Let k and m be positive real numbers such that the function $f(x) = \begin{cases} 3x^2 + k\sqrt{x+1}, & 0 < x < 1 \\ mx^2 + k^2, & x \geq 1 \end{cases}$ is differentiable for all x > 0. Then $\frac{8f'(8)}{f'\left(\frac{1}{8}\right)}$ is equal to _____. **[JEE(Main)April,2023]**

माना धनात्मक वास्तविक संख्याएँ k तथा m इस प्रकार हैं कि फलन $f(x) = \begin{cases} 3x^2 + k\sqrt{x+1}, & 0 < x < 1 \\ mx^2 + k^2, & x \geq 1 \end{cases}$ सभी x > 0 के लिए अवकलनीय है। तो $\frac{8f'(8)}{f'\left(\frac{1}{8}\right)}$ बराबर है : **[JEE(Main)April,2023]**

Ans. 309

Sol. $f\left(1^-\right) = f(1) = f\left(1^+\right)$

$3 + k\sqrt{2} = m + k^2 \quad _____(1)$

$f'\left(1^-\right) = f^1\left(1^+\right)$

$6 + \frac{k}{2\sqrt{2}} = 2m \quad _____(2)$

from (1) and (2)

$3 + k\sqrt{2} = 3 + \frac{k}{4\sqrt{2}} + k^2 \quad \Rightarrow \quad k^2 + k\left(\frac{1}{4\sqrt{2}} - \sqrt{2}\right) = 0$

$k = 0, k = \frac{7}{4\sqrt{2}} \Rightarrow \quad \text{If } k = \frac{7}{4\sqrt{2}}, m = 3 + \frac{7}{32} = \frac{103}{32}$

$$f'(x)=\begin{cases}6x+\dfrac{k}{2\sqrt{x+1}}\\ 2mx\end{cases}$$

$$f'(8)=\frac{103}{2}$$

$$f'\left(\frac{1}{8}\right)=\frac{6}{8}+\frac{k\cdot 2\sqrt{2}}{2\cdot 3}=\frac{3}{4}+\frac{\sqrt{2}\cdot k}{3}=\frac{4}{3} \qquad \text{now} \quad \frac{8\cdot f'(8)}{f'\left(\frac{1}{8}\right)}=309$$

10. Let $f:(-2,2)\to\mathbb{R}$ be defined by

$f(x)=\begin{cases}x[x] & , \quad -2<x<0\\ (x-1)[x], & 0\le x<2\end{cases}$ where [x] denotes the greatest integer function. If m and n respectively are the points in (–2, 2) at which y = |f(x)| is not continuous and not differentiable, then m + n is equal to _____.

[JEE(Main)April,2023]

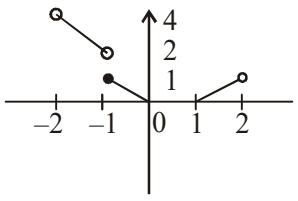
माना $f:(-2,2)\to\mathbb{R}$

$$f(x)=\begin{cases}x[x] & , \quad -2<x<0\\ (x-1)[x], & 0\le x<2\end{cases}$$

द्वारा परिभाषित है, जहाँ [x] महत्तम पूर्णांक फलन है। यदि अंतराल (–2, 2) में उन बिंदुओं, जिन पर y = |f(x)| सतत नहीं है तथा अवकलनीय नहीं है, की संख्या क्रमशः m तथा n है, तो m + n बराबर है _____ **[JEE(Main)April,2023]**

Ans. 4

Sol. $f(x)=\begin{cases}x[x], & -2<x<0\\ (x-1)[x], & 0\le x<2\end{cases}$



|f(x)| = Remain same

m = 1, n = 3

m + n = 4

11. Let [x] be the greatest integer ≤ x. Then the number of points in the interval (–2, 1), where the function $f(x)=|[x]|+\sqrt{x-[x]}$ is discontinuous, is _____. **[JEE(Main)April,2023]**

माना [x] महत्तम पूर्णांक ≤ x है। तो अंतराल (–2, 1) में उन बिंदुओं, जहाँ फलन $f(x)=|[x]|+\sqrt{x-[x]}$ असतत है, की संख्या है _____

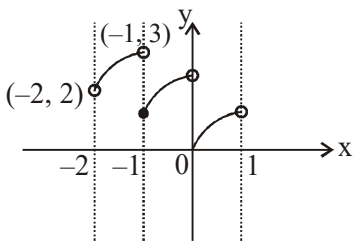
[JEE(Main)April,2023]

Ans. 2

Sol. $f(x)=\left|[x]\right|+\sqrt{x-[x]}$

$x-[x]\geq 0 \Rightarrow x\in R$

$$f(x)=\begin{cases}2+\sqrt{x+2}, & -2<x<-1\\ 1+\sqrt{x+1}, & -1\leq x<0\\ \sqrt{x}, & 0\leq x<1\end{cases}$$



f(x) is discontinuous at two points x = {–1, 0}

12. Consider the function.

$$f(x)=\begin{cases}\dfrac{a\left(7x-12-x^2\right)}{b\left|x^2-7x+12\right|}, & x<3\\ 2^{\frac{\sin(x-3)}{x-[x]}}, & x>3\\ b, & x=3\end{cases}$$

where [x] denotes the greatest integer less than or equal to x. If S denotes the set of all ordered pairs (a, b) such that f(x) is continuous at x = 3, then the number of elements in S is : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) Infinitely many (D) 4

फलन

$$f(x)=\begin{cases}\dfrac{a\left(7x-12-x^2\right)}{b\left|x^2-7x+12\right|} & , \quad x<3 \\ 2^{\frac{\sin(x-3)}{x-[x]}} & , \quad x>3 \\ b & , \quad x=3\end{cases}$$

जहाँ [x] महत्तम पूर्णांक फलन है, का विचार कीजिए। यदि सभी क्रमित युग्मों (a, b), जिनके लिए x = 3 पर f(x) संतत है, का समुच्चय S है, तो में अवयवों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) अनन्त (D) 4

Ans. A

Sol. $f\left(3^-\right)=\dfrac{a}{b}\dfrac{\left(7x-12-x^2\right)}{\left|x^2-7x+12\right|}$ (for f(x) to be cont.)

$$\Rightarrow f\left(3^-\right)=\frac{-a}{b}\frac{(x-3)(x-4)}{(x-3)(x-4)};x<3\Rightarrow\frac{-a}{b}$$

Hence $f\left(3^-\right)=\dfrac{-a}{b}$

Then $f\left(3^+\right)=2^{\lim\limits_{x\to3^+}\left(\frac{\sin(x-3)}{x-3}\right)}=2$ and

f(3) = b.

Hence $f(3)=f\left(3^+\right)=f\left(3^-\right)$

$$\Rightarrow b=2=-\frac{a}{b}$$

b = 2, a = –4

Hence only 1 ordered pair (–4, 2)

13. Consider the function f : (0, 2) → R defined by $f(x)=\dfrac{x}{2}+\dfrac{2}{x}$ and the function g(x) defined by

$$g(x)=\begin{cases}\min(f(t)), & 0<t\le x \text{ and } 0<x\le1 \\ \dfrac{3}{2}+x, & 1<x<2\end{cases}$$. Then : **[JEE(Main)Jan.2024]**

(A) g is continuous but not differentiable at x=1 (B) g is continuous and differentiable for all $x \in (0, 2)$

(C) g is not continuous for all $x \in (0, 2)$ (D) g is neither continuous nor differentiable at x = 1

$f(x) = \frac{x}{2} + \frac{2}{x}$ द्वारा परिभाषित फलन $f : (0, 2) \to R$ तथा $g(x) = \begin{cases} \min(f(t)), & 0 < t \le x \text{ and } 0 < x \le 1 \\ \frac{3}{2} + x, & 1 < x < 2 \end{cases}$ द्वारा परिभाषित

फलन g(x) का विचार कीजिए। तो : **[JEE(Main)Jan.2024]**

(A) x = 1 पर g संतत है परन्तु अवकलनीय नहीं है। (B) सभी $x \in (0, 2)$ के लिए g संतत तथा अवकलनीय है।

(C) सभी $x \in (0, 2)$ के लिए g संतत नहीं है। (D) x = 1 पर g न तो संतत है न ही अवकलनीय है।

Ans. A

Sol. $f(x) = \frac{x}{2} + \frac{2}{x}$

$\Rightarrow f'(x) = \frac{1}{2} - \frac{2}{x^2} = \left(\frac{x^2 - 4}{2x^2}\right)$ for $x \in (0, 2)$

$\Rightarrow f(x) < 0$ $\Rightarrow f(x)$ is decreasing

$$g(x) = \begin{cases} \min\{f(t)\} & ,0 < t \le x \text{ and } 0 < x \le 1 \\ \frac{3}{2} + x & ,1 < x < 2 \end{cases}$$

Since f(x) is decreasing, f(t) is minimum for x in $t \in (0, x] \Rightarrow \min\{f(t)\} = f(x)$

$$\Rightarrow g(x) = \begin{cases} f(x) & , \quad x \in (0,1] \\ \frac{3}{2} + x & , \quad 1 < x < 2 \end{cases}$$

$\Rightarrow g(1) = \frac{1}{2} + \frac{2}{1} = \frac{5}{2}$

Also $g(1^+) = \frac{3}{2} + 1 = \frac{5}{2}$

$\Rightarrow$ g(x) is continuous in (0, 2)

$$g'(x) = \begin{cases} f'(x) & , \quad x \in (0,1] \\ 1 & , \quad x \in (1,2) \end{cases}$$

$g'(1) = f'(1) = \frac{1-4}{2} = \frac{-3}{2}$

$\Rightarrow g'(1) \ne g'(1^+) \Rightarrow$ not differentiable at x = 1

14. Let $f : R - \{0\} \to R$ be a function satisfying $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then :

[JEE(Main)Jan.2024]

माना एक फलन $f : R - \{0\} \to R$ सभी $x, y, f(y) \neq 0$ के लिए $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ को संतुष्ट करता है। यदि $f'(1) = 2024$ है, तो:

[JEE(Main)Jan.2024]

(A) $xf'(x) + f(x) = 2024$ (B) $xf'(x) - 2024f(x) = 0$

(C) $xf'(x) + 2024f(x) = 0$ (D) $xf'(x) - 2023f(x) = 0$

Ans. B

Sol. $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ $f'(1) = 2024$

$f(1) = 1$

Partially differentiating w.r.t. x

$$f'\left(\frac{x}{y}\right)\cdot\frac{1}{y} = \frac{1}{f(y)}f'(x)$$

$y \to x$

$$f'(1)\cdot\frac{1}{x} = \frac{f'(x)}{f(x)}$$

$2024f(x) = xf'(x) \Rightarrow xf'(x) - 2024f(x) = 0$

15. Consider the function $f : (0, \infty) \to R$ defined by $f(x) = e^{-|\log_e x|}$. If m and n be respectively the number of points at which f is not continuous and f is not differentiable, then m + n is : **[JEE(Main)Jan.2024]**

$f(x) = e^{-|\log_e x|}$ द्वारा परिभाषित फलन $f : (0, \infty) \to R$ का विचार कीजिए। यदि उन बिंदुओं की संख्या, जहाँ f संतत नहीं है तथा अवकलनीय नहीं हैं क्रमशः m तथा n है, तो m + n बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 3 (C) 0 (D) 1

Ans. D

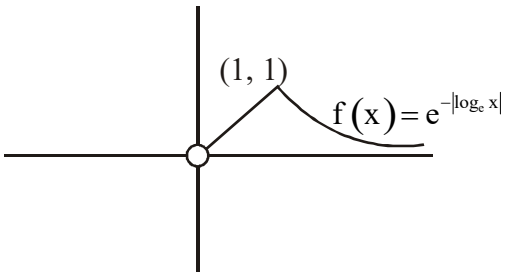
Sol. $f(x) = e^{-|\ln x|}$

$f : (0, \infty) \to R$

$f(x) = e^{-|\log_e x|}$

$$f(x)=\frac{1}{e^{|\ln x|}}=\begin{cases}\frac{1}{e^{-\ln x}}, & 0<x<1\\ \frac{1}{e^{\ln x}}, & x\geq 1\end{cases}$$

$$=\begin{cases}x, & 0<x<1\\ \frac{1}{x}, & x\geq 1\end{cases}$$



f(x) is continuous in $(0, \infty) \Rightarrow m = 0$

from graph, f(x) is not differentiable at 1 point at $x = 1$

$\Rightarrow n = 1$

therefore $m + n = 1$

16. Let $f: R \to R$ be defined as

$$f(x)=\begin{cases}\frac{a-b\cos 2x}{x^2} & ; \quad x<0\\ x^2+cx+2 & ; \quad 0\leq x\leq 1\\ 2x+1 & ; \quad x>1\end{cases}$$

If f is continuous everywhere in R and m is the number of points where f is NOT differential then $m + a + b + c$ equals : **[JEE(Main)Jan.2024]**

माना $f: R \to R$

$$f(x)=\begin{cases}\frac{a-b\cos 2x}{x^2} & ; \quad x<0\\ x^2+cx+2 & ; \quad 0\leq x\leq 1\\ 2x+1 & ; \quad x>1\end{cases}$$

द्वारा परिभाषित है। यदि R के प्रत्येक बिंदु पर f संतत है तथा बिंदुओं की संख्या, जहाँ पर f अवकलनीय नहीं है, m है, तो $m + a + b + c$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 4 (C) 1 (D) 3

Ans. A

Sol. f(x) is continuous $\forall\, x \in R$

$$\Rightarrow f(0) = f\left(0^-\right)$$

$$\Rightarrow 2 = \lim_{x\to 0^-} \frac{a - b\cos 2x}{x^2}$$

$$\Rightarrow a - b = 0 \Rightarrow a = b$$

$$\lim_{x\to 0^-} \frac{2b\sin 2x}{2x} = 2b = 2 \Rightarrow b = 1 = a$$

Also $f(1) = f(1^+)$

$$\Rightarrow 1 + c + 2 = 3$$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = \begin{cases} \dfrac{1-\cos 2x}{x^2}, & x < 0 \\ x^2 + 2, & x \in [0,1] \\ 2x + 1, & x > 1 \end{cases}$$

$$f'(x) = \begin{cases} \dfrac{d}{dx}\left[2\left(\dfrac{\sin x}{x}\right)^2\right] & x < 0 \\ 2x & x \in [0,1] \\ 2 & x > 1 \end{cases}$$

Differentiability at $x = 1$ holds

$$\Rightarrow m = 0$$

$$m + a + b + c = 0 + 1 + 1 + 0 = 2$$

17. Let $f(x) = \left|2x^2 + 5|x| - 3\right|, x \in R$. If m and n denote the number of points where f is not continuous and not differentiable respectively, then $m + n$ is equal to : **[JEE(Main)Jan.2024]**

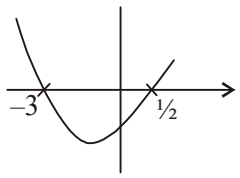
माना $f(x) = \left|2x^2 + 5|x| - 3\right|, x \in R$ हैं। यदि m तथा n क्रमशः उन बिंदुओं, जहाँ पर f संतत नहीं है तथा अवकलनीय नहीं है की

संख्या हैं, तो m + n बराबर है : **[JEE(Main)Jan.2024]**

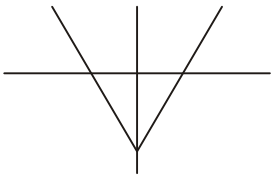
(A) 5 (B) 0 (C) 3 (D) 2

Ans. C

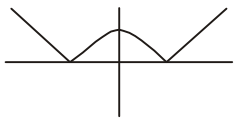
Sol. $y = 2x^2 + 5x - 3$



$y = 2x^2 + 5|x| - 3$



$y = |2x^2 + 5|x| - 3|$



$m = 0, n = 3$

$\Rightarrow m + n = 3$

18. Let for a differentiable function $f: (0, \infty) \to R$, $f(x) - f(y) \geq \log_e\left(\frac{x}{y}\right) + x - y$, $\forall x, y \in (0, \infty)$. Then $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to _____. **[JEE(Main)Jan.2024]**

माना एक अवकलनीय् फलन $f: (0, \infty) \to R$ के लिए, $f(x) - f(y) \geq \log_e\left(\frac{x}{y}\right) + x - y$, $\forall x, y \in (0, \infty)$ है। तो $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 2890

Sol. $f(x) - f(y) \geq \ln x - \ln y + x - y$

$$\frac{f(x) - f(y)}{x - y} \geq \frac{\ln x - \ln y}{x - y} + 1$$

Let $x > y$

$$\lim_{y \to x} f'\left(x^-\right) \geq \frac{1}{x} + 1 \qquad \text{_____(1)}$$

Let $x < y$

$$\lim_{y \to x} f'\left(x^{+}\right) \le \frac{1}{x} + 1 \qquad \text{_____(2)}$$

$$f^{1}\left(x^{-}\right) = f^{1}\left(x^{+}\right)$$

$$f^{1}(x) = \frac{1}{x} + 1$$

$$f'\left(\frac{1}{x^2}\right) = x^2 + 1$$

$$\sum_{x=1}^{20}\left(x^2 + 1\right) = \sum_{x-1}^{20} x^2 + 20$$

$$= \frac{20 \times 21 \times 41}{6} + 20$$

$$= 2890$$

19. If the function $f(x) = \begin{cases} \frac{1}{|x|}, & |x| \ge 2 \\ ax^2 + 2b, & |x| < 2 \end{cases}$ is differentiable on R, then 48 (a + b) is equal to _____.

[JEE(Main)Jan.2024]

यदि फलन $f(x) = \begin{cases} \frac{1}{|x|}, & |x| \ge 2 \\ ax^2 + 2b, & |x| < 2 \end{cases}$ R पर अवकलनीय है, तो 48 (a + b) बराबर _____ है।

[JEE(Main)Jan.2024]

Ans. 15

Sol. $f\left(x\right) = \begin{cases} \frac{1}{x} & x \ge 2 \\ -\frac{1}{x} & x < -2 \\ ax^2 & -2 < x < 2 \end{cases}$

$$f'(x)=\begin{cases}-\dfrac{1}{x^2} & x\geq 2\\ \dfrac{1}{x^2} & x<-2\\ 2ax & -2<x<2\end{cases}$$

at $x=2$

$$-\frac{1}{2^2}=2a\times 2$$

$$a=-\frac{1}{16}\ \&\ \text{continuous also}$$

$$4a+2b=\frac{1}{2}$$

$$-\frac{1}{4}+2b=\frac{1}{2} \qquad b=\frac{3}{8}$$

$48(a+b)$

$$48\left(\frac{-1}{16}+\frac{3}{8}\right)$$

$$48\left(\frac{-1+6}{16}\right)$$

$=15$

20. Let $f:R\to R$ be a function given by

$$f(x)=\begin{cases}\dfrac{1-\cos 2x}{x^2}, & x<0\\ \alpha, & x=0\\ \dfrac{\beta\sqrt{1-\cos x}}{x}, & x>0\end{cases}$$

where $\alpha,\beta\in R$. If f is continuous at $x=0$, then $\alpha^2+\beta^2$ is equal to : **[JEE(Main)April,2024]**

माना एक फलन $f:R\to R$ को

$$f(x)=\begin{cases}\dfrac{1-\cos 2x}{x^2}, & x<0\\ \alpha, & x=0\\ \dfrac{\beta\sqrt{1-\cos x}}{x}, & x>0\end{cases}$$

से परिभाषित किया जाता है, जहाँ $\alpha, \beta \in R$, यदि $x = 0$ पर फलन f संतत है, तो $\alpha^2 + \beta^2$ के बराबर है :

[JEE(Main)April,2024]

(A) 6 (B) 48 (C) 3 (D) 12

Ans. D

Sol. $f\left(0^-\right)=\lim\limits_{x\to 0^-}\dfrac{2\sin^2 x}{x^2}=2=\alpha$

$$f\left(0^+\right)=\lim_{x\to 0^+}\beta\times\sqrt{2}\frac{\sin\frac{x}{2}}{2\frac{x}{2}}=\frac{\beta}{\sqrt{2}}=2$$

$\Rightarrow \beta = 2\sqrt{2}$

$\alpha^2+\beta^2=4+8=12$

21. If the function $f(x)=\begin{cases}\dfrac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}, & x\neq 0\\ a\log_e 2\log_e 3, & x=0\end{cases}$ is continuous at $x=0$, then the value of a^2 is equal to :

[JEE(Main)April,2024]

यदि फलन $f(x)=\begin{cases}\dfrac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}, & x\neq 0\\ a\log_e 2\log_e 3, & x=0\end{cases}$

$x = 0$ पर संतत है, तो a^2 का मान बराबर है : **[JEE(Main)April,2024]**

(A) 1152 (B) 1250 (C) 746 (D) 968

Ans. A

Sol. $f\left(0^+\right)=\lim\limits_{x\to 0}\dfrac{72^h-9^h-8^h+1}{1-\cos h}\times 2\sqrt{2}$

$$=\lim_{x\to 0}\frac{\left(9^h-1\right)\left(8^h-1\right)}{h^2\times\frac{1}{2}}\times 2\sqrt{2}$$

$\ell n9 \times \ell n8 \times 4\sqrt{2}$

$= 24\sqrt{2}\ell n2 \times \ell n3$

22. If the function $f(x) = \dfrac{\sin 3x + \alpha \sin x - \beta \cos 3x}{x^3}$, $x \in \mathbf{R}$, is continuous at $x = 0$, then $f(0)$ is equal to :

[JEE(Main)April,2024]

यदि फलन $f(x) = \dfrac{\sin 3x + \alpha \sin x - \beta \cos 3x}{x^3}$, $x \in \mathbf{R}$, बिन्दु $x = 0$ पर संतत है $f(0)$ बराबर है :

[JEE(Main)April,2024]

(A) 2 (B) 4 (C) –2 (D) – 4

Ans. D

Sol. $f(x) = \dfrac{\sin 3x + \alpha \sin x - \beta \cos 3x}{x^3}$

Continuity at $x = 0$

$$\lim_{x \to 0} = \frac{\sin 3x + \alpha \sin x - \beta(\cos 3x)}{x^3} = f(0)$$

$$\frac{\lim\limits_{x \to 0}\left[3x - \dfrac{(3x)^3}{3!} \ldots.\right] + \alpha\left(x - \dfrac{x^3}{3!} + \ldots\right) - \beta\left(1 - \dfrac{(3x)^2}{2!} + \ldots\right)}{x^3}$$

$$\lim_{x \to 0} \frac{(-\beta) + [3 + \alpha]x + \dfrac{9}{2}\beta x^2 + \left(\dfrac{-27}{6} - \dfrac{\alpha}{6}\right)x^3 + \ldots.}{x^3}$$

For continuous $\beta = 0$, $3 + \alpha = 0 \Rightarrow \alpha = -3$

$$f(0) = -\frac{27}{6} - \frac{\alpha}{6} = \frac{-27 + 3}{6} = -4$$

23. Let $f : [-1, 2] \to R$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t. The number of points, where f is not continuous is : **[JEE(Main)April,2024]**

माना फलन $f : [-1, 2] \to R$, $f(x) = 2x^2 + x + [x^2] - [x]$ द्वारा दिया गया है, जहाँ $[t]$ महत्तम पूर्णांक $\le t$ हैं। उन बिन्दुओं जिन पर f संतत नहीं है, की संख्या है) : **[JEE(Main)April,2024]**

(A) 3 (B) 5 (C) 6 (D) 4

Ans. D

Sol. Doubtful points : $-1, 0, 1, \sqrt{2}, \sqrt{3}, 2$

at $x = \sqrt{2}, \sqrt{3}$

$f(x) = (2x^2 + x - [x]) + [x^2]$ =Discontinuous

↓ ↓

Cont. Discont.

at $x = -1$:

$$\left.\begin{aligned} \text{RHL} \Rightarrow f(x) &= (2 - 1 - (-1)) + 0 = 2 \\ F(-1) &= 2 - 1 - (-1) + 1 = 3 \end{aligned}\right\} \text{Dis.}$$

at $x = 2$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow f(x) 8 + 2 - 1 + 3 &= 12 \\ F(2) = 8 + 2 - 2 + 4 &= 12 \end{aligned}\right\} \text{Cont.}$$

at $x = 0$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow 0 + 0 - (-1) + 0 &= 1 \\ F(0) = 0 \end{aligned}\right\} \text{Dis.}$$

at $x = 1$:

$$\left.\begin{aligned} \text{LHL} \Rightarrow 2 + 1 - 0 + 0 &= 3 \\ F(1) = 3 - 1 + 1 &= 3 \\ \text{RHL} \Rightarrow 2 + 1 - 1 + 1 &= 3 \end{aligned}\right\} \text{Cont.}$$

24. For $a, b > 0$, let $f(x) = \begin{cases} \dfrac{\tan((a+1)x) + b\tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \dfrac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}}, & x > 0 \end{cases}$ be a continuous function at $x = 0$. Then $\dfrac{b}{a}$ is equal to:

[JEE(Main)April,2024]

$a, b > 0$ के लिए, माना $f(x) = \begin{cases} \dfrac{\tan((a+1)x) + b\tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \dfrac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}}, & x > 0 \end{cases}$, $x = 0$ पर एक संतत फलन है। तो $\dfrac{b}{a}$ बराबर है :

[JEE(Main)April,2024]

(A) 5 (B) 6 (C) 4 (D) 8

Ans. B

Sol. $\lim\limits_{x\to 0} f(x) = f(0) = 3$

$$= \lim_{x\to 0^+} \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}} = 3$$

$$= \lim_{x\to 0^+} \frac{ax + b^2x^2 - ax}{b\sqrt{a}x^{\frac{3}{2}}\left(\sqrt{ax + b^2x^2} + \sqrt{ax}\right)}$$

$$= \lim_{x\to 0^+} \frac{b^2}{b\sqrt{a}\left(\sqrt{a + b^2x} + \sqrt{a}\right)}$$

$$= \frac{b}{\sqrt{a}\cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \qquad ...(1)$$

$$= \lim_{x\to 0^-} \frac{\tan(a+1)x + b\tan x}{x}$$

$$= \lim_{x\to 0^-} \frac{\frac{\tan(a+1)x}{(a+1)x}(a+1)x + \frac{b\tan x}{x}\cdot x}{x}$$

$= a + 1 + b = 3 \Rightarrow a + b = 2$ ...(2)

from (1) and (2)

$a + 6a = 2$

$$a = \frac{2}{7} \qquad b = \frac{12}{7}$$

25. Let [t] denote the greatest integer less than or equal to t. Let $f : [0, \infty) \to R$ be a function defined by $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$. Let S be the set of all points in the interval [0, 8] at which f is not continuous. Then $\sum\limits_{a\in S} a$ is equal to _____. **[JEE(Main)April,2024]**

माना [t], t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। माना फलन $f : [0, \infty) \to R$, $f(x) = \left[\frac{x}{2} + 3\right] - \left[\sqrt{x}\right]$ द्वारा परिभाषित है। माना S, अन्तराल [0, 8] में उन सभी बिन्दुओं का समुच्चय है, जहाँ f संतत नहीं है। तो $\sum\limits_{a\in S} a$ बराबर _____ है।

[JEE(Main)April,2024]

Ans. 17

Sol. $f(x)=\left[\frac{x}{2}+3\right]-\left[\sqrt{x}\right]$

$=\left[\frac{x}{2}\right]-\left[\sqrt{x}\right]+3 \qquad \because \in[0,8], \frac{x}{2}\in[0,4], \sqrt{x}\in\left[0,2\sqrt{2}\right]$

Point of discontinuity

When $\frac{x}{2}$ is integer & $\sqrt{x}$ is int.

$\frac{x}{2}=0,1,2,3,4 \qquad \sqrt{x}=0,1,4,9$

$x=\underset{\substack{\downarrow\\ \text{Right}\\ \text{cont}}}{0}, \underbrace{2,4,6,8}_{\text{discont}} \qquad x=\underset{\substack{\downarrow\\ \text{cont}}}{0}, \underbrace{1,4,9}_{\text{discont}}$

$\therefore$ f(x) is discontinuous at x = 1,2,6,8

Check continuity at x = 4

$f\left(4^{+}\right)=\left[2^{+}\right]-\left[\sqrt{4^{+}}\right]+3=3$

$f\left(4^{-}\right)=\left[2^{-}\right]-\left[\sqrt{4^{-}}\right]+3=1-1+3=3$

$f\left(4\right)=\left[2\right]-\left[\sqrt{4}\right]+3=3$

$\therefore$ f(x) is continuous at x = 4

Point of discontinuity = 1, 2, 6, 8

$\sum_{a\in s} a=1+2+6+8=17$

26. Let $f:(0,\pi)\to R$ be a function given by $f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8, & x=\frac{\pi}{2}\\ \left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$ where a, b $\in$ Z. If f is continuous at $x=\frac{\pi}{2}$, then a^2+b^2 is equal to_____. **[JEE(Main)April,2024]**

माना $f:(0,\pi)\to R$ एक फलन दिया गया है जो कि $f(x)=\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2}\\ a-8, & x=\frac{\pi}{2}\\ \left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi\end{cases}$ है, जहाँ $a, b\in Z$ है। यदि $x=\frac{\pi}{2}$ पर संतत है, तो a^2+b^2 is बराबर _____ है। **[JEE(Main)April,2024]**

Ans. 81

Sol. As f(x) is continuous at x

$$\Rightarrow Lf\left(\frac{\pi}{2}\right)=f\left(\frac{\pi}{2}\right)=Rf\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \lim_{x\to\frac{\pi}{2}}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}=a-8=\lim_{x\to\frac{\pi}{2}}\left(1+|\cot x|\right)^{\frac{b}{a}|\tan x|}$$

$$\Rightarrow\left(\frac{8}{7}\right)^{0}=a-8=e^{\frac{b}{a}}$$

$\Rightarrow a = 9$ and $b = 0$

Hence $a^2 + b^2 = 81$